STAGE 1: Identify Thesis Supervisors (supervisory committee)

1) Students must select their potential MSc thesis supervisors (Lead- & Co-supervisors) within the first 8 weeks of the Program's second semester and inform the MSc Program Coordinator, who is going to initiate the required approval of your supervisory committee by the departmental MSc Program Committee. **Both supervisors must be from the department**. In exceptional cases, with the approval of the Department MSc Committee, HoD or Dean, it will be accepted to have the Co-Supervisor external to the Department in which the student is enrolled. But, with no exception, the Lead Supervisor must be a faculty of the Department of the student. In the case of having an external member as Co-Supervisor, that member will act as internal to the effects of evaluation of the Master Thesis. Furthermore, an external examiner needs to be assigned to each student, who will not be part of the student supervisory committee but will be involved in the approval of the final MSc thesis report and the evaluation of the MSc. Thesis defense. The external examiner must be external to your department and can be an academic from another NU department, university or, alternatively, an expert from the industry holding an appropriate academic degree and specializing in your thesis' scientific field. Department's MSc committee will choose the External Examiner from a list of supervisor-proposed candidates. In all stages, all involved individuals and bodies are responsible for identifying and declaring potential or perceived conflict of interest among involved parties and following the rules and guidelines mentioned in Appendix VI.

Constructive supervision is a significant component aiming in the success of your thesis work and requires the vivid interaction between you and your supervisors. However, you should never forget that it is you, the student, who is carrying out the work and it's your motivation, academic knowledge, and interest central in making the supervision process work.

Your supervisory committee comprises academic professionals that will help you track appropriate research sources and support your research and the compilation of your thesis work. Your supervisor may also refer you to other experts (either internal or external to the university) who may have specialized knowledge in the specific topic of your thesis. Your Lead Supervisor is responsible for ensuring that the Master's thesis meets the goals and requirements set by the School of Engineering and Digital Sciences. Your supervisory committee will be able to predict common pitfalls and protect you from them while at the same time provide you with advice helping you meet your thesis' objectives. However, keep in mind that it is required by you to be the dynamic party in the interaction with your supervisors. Discussion and critical argumentation are key features that should be present in your meetings. Supervision is not a monologue, and neither the supervisors nor the student should restrain themselves from asking the other party for clarification of claims. Mutually challenging dialogues can help you in delivering a successful result.

STAGE 2: Selection of topic

Supervisors are in position to suggest appropriate MSc thesis' topics. These may stem from research work being conducted at the school/department or may arise from material covered during your coursework. Furthermore, thesis' topics may be also related to work carried out in the context of research projects involving industrial partners. The topic of the thesis is decided in discussions between you and your chosen supervisors; however, the final choice is always made by you. In this regard, it is customary for interested department faculty members to announce topics and/or areas of interest in helping you choose your thesis topic.

Furthermore, during your first supervised meetings you should spend some time discussing, in addition to academic matters, all practical matters that may arise during your work:

- Do you need all-round supervision? Have you got prior experience in using equipment or software required in the proposed topic? Will you be able to use them?
- When can you meet with your supervisor? How often? Can you set up regular meeting intervals or do you prefer an on-demand approach? Keep in mind that supervisors are bound to offer at least 12 hours of supervision for a 48-credit master thesis.
- How ambitious are you? Are you thinking of later applying for a funded PhD position?

Finally, a *Supervision Agreement* form, found in Appendix II, must be filed in, signed, and submitted to the MSc Coordinator for review and approval by MSc Committee and Head of Department by the end of the second semester. The Supervision Agreement must state the proposed thesis title, supervisors, start date, and intended submission date.

STAGE 3: Submission of your thesis proposal

When you and your supervisor come to an agreement for an appropriate thesis topic, you are required to submit a thesis proposal/candidature within the first four (4) weeks of the Program's 3rd semester. This proposal will be presented with the aid of a short report and defended orally in front of an academic panel, comprising your two supervisors and a third faculty from the Department of Electrical and Computer Engineering appointed by the MSc ECE Committee and approved by the Head of Department. This panel will decide on the appropriateness of your proposed thesis topic and the scientific concreteness of the methodologies you are aiming to apply. Your research/thesis topic proposal should clearly address the following items:

- Outline of the problem/area of application
    - Explain why you think it is worth investigating
    - Set your ideas into a theoretical/academic context
- Aims and Objectives
    - Describe what you are aiming to achieve
    - Present the steps and approaches you will employ for reaching your goals
- Methodology
    - Explain what methods you intend to use when researching and developing your work
    - Use a descriptive writing approach corresponding to the detail required for the panel's comprehension of your approach.
- Scope and constraints
    - Set clearly your scope and anticipated constraints:
        - Your selected topic may be vast with numerous applications and thus, you might want to limit your work in a particular area of application
        - You may not be able to conduct some research due to constraints on time, cost, or availability of resources
- Discuss requirements on resources
    - Do you need any special lab equipment?
    - Is literature review possible with library's resources?
    - Are any materials and/or consumables required in your research?
- Propose a draft timetable for your thesis

The panel may accept your proposal or provide you with feedback and change suggestions that will help you meet the required academic standards for starting your thesis. If the proposal is deemed unacceptable you will have a second chance to revise and present your modified proposal within four weeks. If

your proposal is not accepted for a second time, you will be recommended for dismissal from the program[1].

Your thesis proposal should be obviously discussed during the preliminary meetings with your supervisory committee. Your Lead Supervisor will usually provide pertinent literature and/or additional resources to accelerate your initial work. Finding suitable and reliable information may prove challenging, but there are many ways including library books, databases, international sources, articles, journals, reviews, and a lot more.

STAGE 4: Carrying out research and preparation of your thesis manuscript

Once your proposal has been officially approved, the actual work may begin. It is crucial that you are always well-prepared in meetings with your supervisor. In this context, it is a good practice to always keep minutes of your meetings and circulate agendas with clearly outlined discussion points and expected results prior to your meetings. This makes it easier for the supervisor to focus on significant issues, leading to a better response for you. If you feel that you may have misunderstood a concept, or you are not certain of the steps required for performing a particular task, ask your supervisor for clarifications or further guidance. The supervisors should always guide you with advices on the topics and tasks you should put emphasis on and at the same time turn you away of meaningless tasks that may waste your time.

Try to establish a communication channel that suits both you and your supervisors. Emailing is an easy, asynchronous way of communication that overcomes time and place barriers. Furthermore, since it is primarily based on writing, it requires a certain amount of prior thinking and planning that helps you avoid getting 'off-track' as it may be the case when speaking. However, emails are cumbersome when lengthy and lack the directness of a real meeting. So, it is important to balance the ways of communication based on your needs and supervisor's availability.

Try to follow the work schedule as close as possible and report unexpected delays or difficulties to your supervisor. This does not mean that whenever you are faced with a difficulty, you will turn to your supervisor for doing the work for you. It essentially means that after putting reasonable effort on accomplishing a specific result which is still elusive, you should turn to your supervisor for additional guidance.

Finally, you must keep in mind that writing a thesis cannot happen in one go. You should, as soon as possible, keep track of your work, make notes and sketches, write intermediate reports so that when your work has approached a certain maturity level, you'll be able to compile, with the aid of this material, a successful thesis' manuscript. **In this regard a progress report is required to be submitted by you to the MSc Coordinator as a proof of progress by the end of the 3rd semester. This progress report should include as a minimum requirement a complete literature review and must be approved by your supervisor.**

STAGE 5: Thesis submission & Defense of your work

Before submitting your manuscript, your supervisor will check it thoroughly and give you feedback on corrections and changes that need to be made. Usually, thesis's revision may take up to 1-2 weeks, and an appropriate amount of time should be also reserved for making corrections.

[1]Failing of MSc thesis or dismissal from program are subject to regular appeal process and rules established in the program handbook and guidelines

When you have prepared the revised document, you submit to your supervisory committee and external examiner for evaluation. Keep also in mind that you should aim at meticulously following your supervisor's comment and corrections so that a series of multiple revisions can be avoided. When your supervisory committee & the external examiner approve the final document, your Lead supervisor will fill the required form and you will get permission to submit your final thesis report for evaluation.

The overall grade for your degree is calculated as the credit-weighted average of all course grades. Additionally, a SUCCESSFUL submission and defense of your thesis is required to be considered for graduation; see §Thesis Grading below for the employed scheme in your thesis evaluation.

Thesis submission process involves the following steps:

- **Thesis report submission (in electronic format).**
- **Thesis oral defense in front of the examination committee.**
- **Thesis revision, if required.**
- **Submission of final version of your report (Thesis manuscript).**
- **Thesis mark appeal, if any. The appeal should be submitted to the MSc ECE committee, which is responsible for the Thesis evaluation process.**

The exact deadlines for each submission process step will be announced every year in due time. Commonly, the submission and examination process begin about 1.5 months before the end of the program's 4th (last) semester and may end few days after the end of the semester.

If your supervisory committee does not approve your thesis for defense or you fail the defense, you may continue to work on your thesis during the immediate summer semester and defend the amended thesis in the summer term before the beginning of the following fall semester. You will not receive a scholarship and you may need to cover the tuition and other fees by yourself during the summer term.

You may also request the Dean's permission to extend your MSc program for the following Fall semester if needed to complete the program requirements. The student's academic supervisor must endorse the request. During this period, you will not receive a scholarship and you may need to cover the tuition and other fees by yourself. If you cannot present and/or defend your thesis during this last semester, you will be recommended for dismissal from the program[2].

In any case, re-examination of the MSc thesis may be permitted only once, with the approval of the Dean of School of Engineering and Digital Sciences.

Thesis Assessment Criteria

The grade assigned depends on the level to which the following criteria have been met:

Manuscript Grading (MSc Thesis Manuscript)

Maximum MSc Thesis Manuscript score: **100**

- ***Presentation of the research problem and thesis' objectives (10%)***
  - Is the research problem clearly specified and contextualized?
  - Are the research questions and hypotheses clearly formulated?
  - Does the thesis capture the relevance, rationale, and objectives of the proposed research?
- ***Literature and technology review (15%)***
  - Does the thesis include a comprehensive review and critical discussion of the relevant literature and/or technological developments?

[2] Failing of MSc thesis or dismissal from program are subject to regular appeal process and rules established in the program handbook and guidelines

- Is there a description on how the conducted research positions itself within the generic context of works which have been published in the area?
- Is the relevant background theory covered? Are the presentation, discussion and explanation provided, adequate? Has the theory been contextualized appropriately within the framework of the research problem being investigated?
- Have the latest theoretical developments in the area been presented and described?
- Does the student demonstrate a systematic understanding of the relevant background material and knowledge?

- ***Methodology, design and implementation (35%)***
  - Are the adopted methodologies and/or design approaches clearly justified and described?
  - Is the implementation well explained?
  - Is there a clear identification of any limitations, assumptions and constraints which affect the application of the employed methodology, design approach and implementation?
- ***Testing, results, analysis, evaluation concluding remarks & future work (30%)***
  - Are the test procedures sound and objective?
  - Do the proposed tests address the research problem being investigated?
  - Are the test conditions, assumptions, constraints, and limitations clearly identified?
  - Are the results clearly presented, analyzed objectively and critically evaluated?
  - Do the concluding remarks summarize the work done? Are there suggestions for any future development and/or enhancements?
- ***Structure and presentation of thesis (10%)***
  - Are the thesis contents well structured, focused, and easy to follow?
  - Are the student's contributions and assumptions clearly communicated to the reader?
  - Is it in compliance with the given guidelines?
  - Is it clearly presented and organized? Is the grammar and usage of English of an appropriate level?

Oral Presentation Grading (MSc Thesis Defense)

Maximum MSc Thesis Defense score (presentation + technical content) **100**

PRESENTATION SCORE: *(Maximum presentation score: 50)*

- ***Speech & Style*** *(10p.)*
  - Clear and easily understood. Correct use of terms.
  - Easy-to-understand sequence. Professional appearance. Use of good English.
- ***Structure of the Presentation*** *(10p.)*
  - Logical sequence, good flow. Supporting body of literature mentioned.
  - Development of topics described clearly. Smooth progression from topic to topic.
  - Key points & challenges sufficiently highlighted.
- ***Layout of Visual Aids*** *(10p.)*
  - Clear power point slides, uncluttered. Concise & precise slides.
  - Use of good English. Good use of charts, tables, diagrams, etc.
- ***Questions & Answers*** *(20p.)*
  - Clearly understood the question.
  - Concise answer responding to the point of the question.

TECHNICAL CONTENT SCORE: *(Maximum technical content score: 50)*

- ***Introduction*** *(10p.)*
  - Problem statement & project objectives. Coverage of all main points of the project.
  - Literature review and conclusions. Relevance to the need of industry, society etc.
- ***Technical Competency*** *(30p.)*
  - Viability of the design concept. Justification of the approach
  - Design methodology. Practical Implications.
  - Quality of the concept presentation. Interpretation of the achieved results.

- Use of relevant tools/equipment/software. Costs and efficiency considerations.

- ***Conclusions, Future Work & Professional ethics*** *(10p.)*
  - Conclusions: advantages and disadvantages.
  - Level of the project objectives achievement.
  - Future work and possible improvements.
  - Consideration in design and solution. Applicability to real-life situations.
  - Compliance with good practices and standards.

Thesis Grading

The MSc. Thesis must be compiled in a report (manuscript) according to the specification provided herein and defended in front of MSc. Examination Thesis committee, which comprises your two supervisors and your external examiner. The MSc. Thesis manuscript and MSc Defense Oral presentation will be evaluated using the assessment criteria and weighting presented in the previous section, Thesis Assessment Criteria and Oral Presentation Assessment Criteria, respectively. The Thesis manuscript evaluation (M) contributes 70% to your final thesis evaluation while the remaining 30% comes for your Oral presentation (O).

Thesis is not graded with a letter scale as in the case of other MSc courses, but a SUCCESSFUL / UNSUCCESSFUL attribution is utilized. For a successful completion of your thesis, you are required to achieve an overall supervisory committee evaluation greater or equal to 75%. The examination committee members' evaluations contribute to your MSc thesis result as follows:

1. External examiner's (EE_M and EE_O) evaluations are weighted with a 30% weight, and
2. Lead and Co-Supervisor's evaluations (S1_M, S1_O & S2_M, S2_O) are weighted with a 35% weight each

Hence, your final thesis evaluation (FE) is calculated as follows:
FE = 0.3 O + 0.7 M,
where O = (0.3 EE_O + 0.35 S1_O + 0.35 S2_O) and M = (0.3 EE_M + 0.35 S1_M + 0.35 S2_M)
If FE is greater or equal to 75%, your thesis manuscript & defense is considered SUCCESSFUL, otherwise it will be considered UNSUCCESSFUL.

In case of a difference larger than 25% between evaluation marks given by External and average of Internal Examiners, the Department MSc Committee will decide the final evaluation of the thesis. The percentage difference will be calculated using the following relation:

$$\text{Percentage Difference} = \frac{|V_i - V_e|}{\frac{(V_i + V_e)}{2}} \times 100,$$

where V_i is the average evaluation of the internal members and V_e the evaluation of the external member.

Manuscript Structure & Formatting

Detailed description of manuscript's structure along with specific guidelines for the document's styling can be found in Appendix III – Manuscript Format Specifications.

Referencing

Whenever writing a piece of academic work, you are required to acknowledge the sources of data and information that you have used. This permits you to:

- prove that your work has a substantial factual basis;
- offer your readers the means to identify and retrieve the references for their own use;
- acknowledge the creators/authors of material/methods you have used/employed in your own research work
- support the research methodology and approaches you have used to reach your conclusions.

You can use any established engineering citation methodology to reference any material used in your work. For more information on ASME[3]'s referring style and alternative approaches see Appendix IV – Referencing style.

Always keep in mind that referencing is divided into two key components:

1. *In-text references* where references might be numbered in the order of appearance, as in [1] or using the author's name and date of publication as in (James et al., 2002);
2. *A reference list* displayed at the end of the piece of work which provides full details of all references cited in-text. The references can be ordered as they appear in text or in alphabetical order according to the selected style. In any case, the identification mark for each item in the list must coincide with the in-text reference used.

[3] THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Appendices

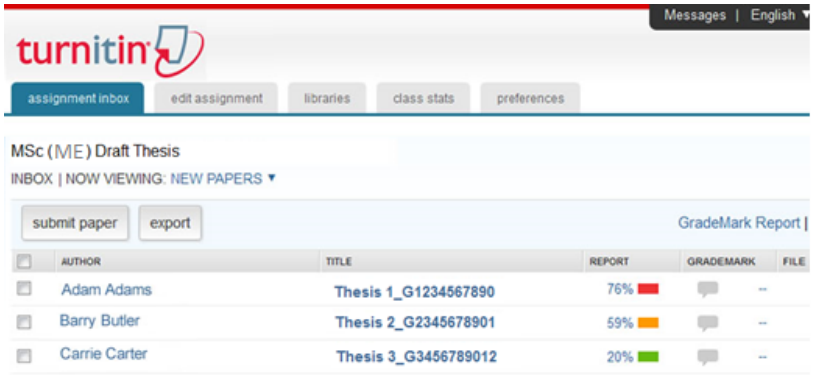
Appendix I - Turnitin

Turnitin is a web-based originality checking service that is used by many universities worldwide to prevent plagiarism. When a student's work is submitted to Turnitin it is matched against millions of internet pages, electronic journals, books, and a database of all previously and concurrently submitted assignments. Turnitin then generates an originality report providing a summary of matching or similar text found in the submitted work. Turnitin can be used to check sources that have been correctly acknowledged and cited.

The higher the percentage indicated on the originality report, the greater the amount of text copied. The similarity percentage index ranges from 0% to 100% and is specified by distinct Colors:

- **Blue** (No matching words)
- **Green** (1 - 24% similarity index)
- **Yellow** (25 – 49% similarity index)
- **Orange** (50 – 74% similarity index)
- **Red** (75 – 100% similarity index)

Acceptable ranges for plagiarism using Turnitin is ideally below 20%. The figure below illustrates a sample originality report.



Appendix II – Supervision Agreement Form

MSc Thesis-Supervision Agreement Form
School of Engineering and Digital Sciences, Nazarbayev University

NAZARBAYEV UNIVERSITY

Department of Electrical and Computer Engineering
53 Kabanbay Batyr Ave. Nur-Sultan, Kazakhstan, 010000
Tel. No: +7 (7172) 70 64 23, 70 65 43, 70 65 99
E-mail address: seng@nu.edu.kz
Web-page: seng.nu.edu.kz

Student Name and ID	Name: ID:		
Program	**Master of Science in Electrical and Computer Engineering (MSc-ECE)**		
Lead Supervisor	Name: Institute / School / Department: Address:		
Co-supervisor	Name: Institute / School / Department: Address:		
External Examiner	Name: Institute / School / Department: Address:		
Proposed thesis title			
Starting date		Intended date of thesis submission	

RIGHTS AND OBLIGATIONS IN THE SUPERVISORY RELATIONSHIP

- The supervisor and the student should, at the outset of the supervisory relationship, discuss and agree on the format of supervision, the expected progress, and the intended date of thesis submission.
- The supervisor and the student should observe the regulations and instructions governing the supervision of the master thesis.
- Regulations and instructions regarding conflict of interest (as described in Appendix VI of MSc ECE handbook) in supervision and assignment of external examiners should be observed by the student, the supervisor, and the MSc ECE committee.

Student

- The student should be well-prepared for meetings with supervisors.
- The student should provide sufficient notice to the supervisors if he/she is not able to attend a scheduled meeting.
- When there are serious problems in supervision, the student should immediately bring this to the attention of the Master's Program Coordinator or Head of Department.
- In case the thesis cannot be completed within the period of the agreement, the student must apply for an extension of the supervision agreement.

Supervisors

- The supervisors should be familiar with and follow the ethical guidelines of supervisors at Nazarbayev University. He/she should also ensure that students are aware of the guidelines.
- The supervisors should ensure that the first meeting with the student takes place shortly after having been assigned, in which the supervisors should discuss with and inform the student about how the supervision is to be organized.
- The supervisors should offer the student at least 12 hours of supervision for a 48-credit master thesis, distributed evenly throughout the supervision period.

- The supervisors should be well prepared for the meetings with the student.
- The supervisors should discuss with the student as well as evaluate the plan, the methodology and execution of the proposed research. He/she should assist the student in planning the research with a view to completing it within the normal period of study.
- The supervisors are expected to read and give detailed comments on the draft of the thesis chapters at least once, but the supervisors will exercise their own judgment in how far it is necessary to give detailed comments on revised chapters and the completed thesis.
- The supervisors may, if appropriate, carry out part of the supervision in small research seminars.
- The supervisors should, through the meetings with the student, monitor the progress of the student's work and evaluate the progress in relation to the planned schedule.
- The supervisors should give sufficient notice to the student if he/she needs to re-schedule a meeting.

Modification or Termination of Supervision Agreement

- Changes can be made to the supervision agreement (i.e. changing the research topic) when both the student and lead supervisor agree.
- Changes to the supervisor and/or co-supervisor and/or research topic can only happen before the end of the program's 2nd semester.
- When the supervisor is to be absent for an extended period of time in the course of the supervisory relationship, the department should, in consultation with the student, determine how the supervision can be organized during this period.
- In case the department considers that the student is not acting according to the supervision agreement, he/she should be notified in writing.
- If the supervisory relationship is not deemed, by the student or the supervisor, to be working satisfactorily, for academic or other reasons, either side may request to be released from the supervisory relationship. A new supervisor should then be arranged.
- The student may be considered as not acting according to the supervision agreement in cases when:
  (a) the student fails to submit the master thesis within the semester which the thesis is due for submission and has not been given approval for an extension and a new submission date.
  (b) the student has not contacted the supervisor for one semester. In such a case, the research can be taken up by other students.
  (c) the allocated hours of supervision have been used up, and the student has no approved plan for submission of thesis.
- The supervision agreement ends when the student:
  (a) submits the master thesis
  (b) is considered as not acting according to the agreement and has been notified of it in writing, and has not received approval for a new plan for submission
  (c) forfeits or gives up the right to his/her place in the master program concerned.

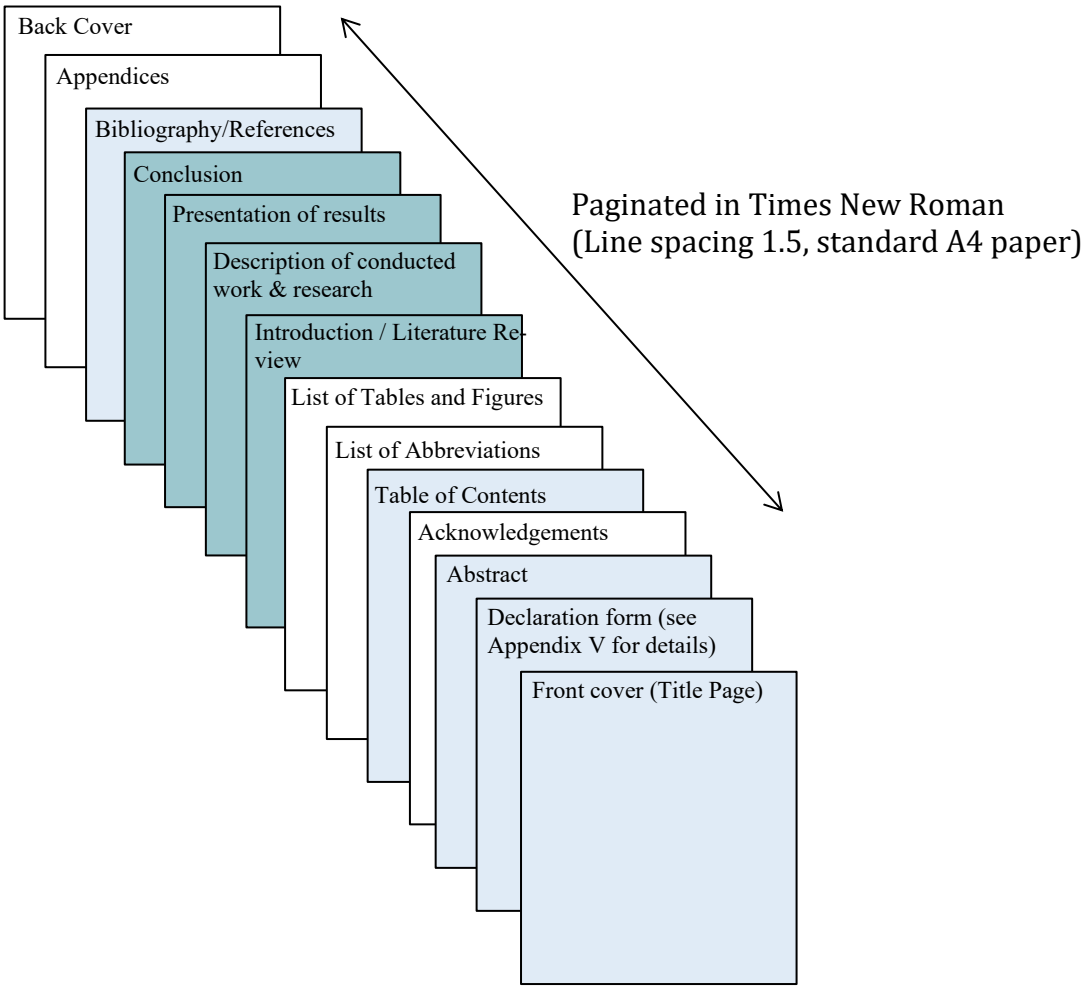
Student	Date	Signature
Lead Supervisor	Date	Signature
Co-supervisor	Date	Signature
Master's Program Coordinator	Date	Signature
Head of Department	Date	Signature
Dean of School	Date	Signature

(Dean's approval is required only for the exceptional case where a supervisor or co-supervisor is external to the department)

Appendix III – Manuscript Format Specifications

1. **Style Requirements**
    - The thesis must be written in English language. Quotations in languages other than English may be included; however, the quotations' translation should be also included.
    - An abstract of the thesis, not exceeding **500** words, should be part of the preliminary material.
    - Supplementary materials, such as questionnaires, large data sets, or copies of photographs, may be placed into appendices. The appendices must be consecutively paginated with the text. The paper quality and margins of the appendices must conform to the standards for the rest of the thesis.
2. **General Thesis Format Requirements**
    - The MSc. Thesis format will be strictly according to the specifications described in this handbook and needs to be limited to **100** pages excluding appendices.
    - The font type will be Times New Roman, in black and size 12 point with the exceptions noted in the detailed guidelines below. The text line-spacing will be double. Lengthy quotations, footnotes, and bibliographies may be single-spaced.
    - Your thesis should be divided into chapters. A chapter should be divided into sections and subsections. Sections and subsections of chapters are to be identified by numbers. Chapters use 1st level numbering, i.e., Chapter 1, 2, 3 etc. chapter sections use 2nd level numbering, i.e., C.1, C.2 where C is the number of the corresponding chapter; e.g., the 3rd section in the 2nd chapter should be numbered as 2.3. Subsections use 3rd level numbering and their numbers are preceded by the chapter and section number, e.g., 2.1.4, 2.1.5 etc. Numbering should be used up to subsections.
    - Computer-generated figures, graphs and other diagrams are acceptable. Each diagram should be numerated, include a caption, and should not be divided between two pages.
    - Tables may be used either in the text or in appendices. All tables should include a numerated header (caption) and named rows and columns.
    - All mathematical and/or chemical equations and relations are considered as text and numbered using a chapter numbering scheme; see above. Detailed, lengthy derivations and mathematical proofs should be placed in Appendices.
    - Page headers or decorative borders should not be used.
    - The text should be justified. The beginning of the first line of text of each paragraph should be indented to **1.25 cm**.
    - The page margins for the text will begin at least **2.5cm** from all sides.
    - The second and subsequent pages should be numbered in Arabic figures in the middle of the top of each page.

3. **Detailed Guidelines & Examples**
Order of manuscript elements and chapters



Required elements/chapters

Optional elements

Required chapters (Additional chapters may also be present)

NOTE: All text should by typed using Times New Roman font face. If your word-processing software does not support Times New Roman, substitute with a font face that closely resembles it.

Pagination: All pages must be numbered; page numbers must be displayed on all pages, except the title page. Numbering should be placed in the middle of the top or bottom of each page

Front Cover

Times New Roman (Bold), Size 16

THESIS TITLE

Times New Roman (Bold), Size 15 Enter your name

Authors Name, Degrees held

Times New Roman, Size 15 Degrees currently attained (i.e. B. Eng)

Times New Roman (Bold), Size 15

Submitted in fulfilment of the requirements for the degree of Master of Science in Electrical and Computer Engineering

NAZARBAYEV UNIVERSITY

Authorized University logo

Times New Roman (Bold), Size 15 School, Department, and University Name

School of Engineering and Digital Sciences
Department of Electrical and Computer Engineering
Nazarbayev University

53 Kabanbay Batyr Avenue,
Nur-Sultan, Kazakhstan, 010000

Times New Roman, Size 15 Institution Address

Supervisors: Supervisors' names

Times New Roman (Bold), Size 15 Date of thesis completion (e.g., December 2016)

Date of Completion

Insert the declaration form (see Appendix V) as an unnumbered separate page following the cover page.

Abstract & Acknowledgements (use the same styling)

2

Times New Roman, Size 12, (Alternatively you may print the page numbers at the bottom of the page)

Times New Roman (Bold), Size 16

Acknowledgements

Times New Roman, Size 12 (If a new page is needed due to many acknowledgments the text will commence at the top of the next page under the header).

Use this section to offer "credit where credit is due". An acknowledgement page is optional, however, most thesis include brief statements of appreciation or recognition of special support. There is no limitation on the number of pages you may use for acknowledgements. An example paragraph could be something in the words of:

Firstly, I would like to express my uttermost gratitude towards my supervisors _____ and _____. It has been their supervision and direction throughout the duration of my studies which has allowed me to successfully complete this Master's program. I am appreciative for all the hours of discussion they have offered me, especially in the areas of communications and embedded systems.

Secondly, I would like to show my indebtedness to the administration staff at Nazarbayev University whose efforts a lot of the time go unnoticed …….

Thirdly, I would like to thank my family/friends/partner …….

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Times New Roman (Bold), Size 16

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Times New Roman, Size 12

Dedicated chapter if literature review is extensive

Times New Roman (Bold), Size 12

List of Abbreviations

5

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List of Abbreviations & Symbols

A	Area of planar quadrilateral element
AijI	Component of velocity normal to body surface
BEM	Boundary Element Method
BIE	Boundary Integral Equation
b	Base of triangle (see Figs …)
CFD	Computational Fluid Dynamics
d	distance
FEM	Finite Element Method
R'	Region exterior to body surface

Times New Roman, Size 12, Left side acronym/symbol, right side definition

List of Figures (List of Tables should follow a similar format)

Times New Roman, Size 12, Header.

Times New Roman (Bold), Size 16

Times New Roman, Size 12

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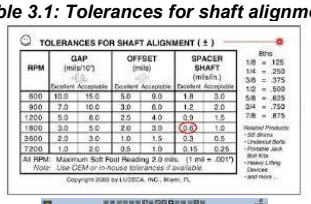
Chapter 3 – Computation of Lifting Flow

The body of context is the substance of your thesis. It should introduce, investigate and verify your findings. The conclusion section is often the last part of writing, summarizing and discussing the overall results.

Times New Roman (Bold, Italic), Size 12, Note: Table captions come before tables. Make sure that font-sizes in tables are suitable for reading.

Table 3.1: Tolerances for shaft alignment



TOLERANCES FOR SHAFT ALIGNMENT (±)

RPM	GAP (mils/10")		OFFSET (mils)		SPACER SHAFT (mils/in.)	
	Excellent	Acceptable	Excellent	Acceptable	Excellent	Acceptable
600	10.0	15.0	5.0	9.0	1.8	3.0
900	7.0	10.0	3.0	6.0	1.2	2.0
1200	5.0	8.0	2.5	4.0	0.9	1.5
1800	3.0	5.0	2.0	3.0	0.6	1.0
3600	2.0	3.0	1.0	1.5	0.3	0.5
7200	1.0	2.0	0.5	1.0	0.15	0.25

At RPM: Maximum Soft Foot Reading 2.0 mils. (1 mil = .001")
Note: Use CEM or in-house tolerances if available.

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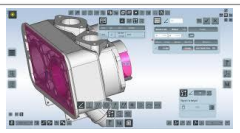
Bins
1/8 = .125
1/4 = .250
3/8 = .375
1/2 = .500
5/8 = .625
3/4 = .750
7/8 = .875

Related Products:
- Sill Shims
- Undercut Bolts
- Portable Jack Bolt Kits
- Heavy Lifting Devices
- and more ...



Times New Roman (Bold, Italic), Size 12, Note: Figure captions are placed below the figure. Make sure all figure elements are clear and readable (Always use vector graphics, if possible).

Figure 3.4: 3d Model of component B

$$\vec{\nabla}\cdot\left(-k\vec{\nabla}T\right)=q_{gen}-\rho c\frac{dT}{dt}$$

$$-k\vec{\nabla}^2T+\rho c\frac{\partial T}{\partial t}=q_{gen}$$

$$\nabla^2T-\frac{1}{\alpha}\frac{\partial T}{\partial t}=-\frac{1}{k}q_{gen} \qquad (3.1)$$

Times New Roman, Size 12, Note: Your document editor should support mathematical symbols and formulas

Bibliography/References

Times New Roman, Size 12, Header.

Times New Roman (Bold), Size 16

Times New Roman, Size 11, Conform to the referencing section of this handbook.

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Bibliography/References

[1] Aftosmis MJ, Melton JE and Berger MJ. Adaptation and surface modeling for Cartesian mesh methods. AIAA Paper, Collection of Technical Papers. Pt. 2 (A95-36501 09-34) 95-1725. In 12th AIAA Computational Fluid Dynamics Conference and Open forum, San Diego, 1995.

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[3] Allmaras S. Analysis of a local matrix preconditioner for the 2-D Navier–Stokes equations. AIAA Paper 93-3330. In AIAA 11th Computational Fluid Dynamics Conference, Orlando, 1993.

[4] Allmaras S. Analysis of semi-implicit preconditioners for multigrid solution of the 2-D Navier–Stokes equations. AIAA Paper 95-1651. In AIAA 12th Computational Fluid Dynamics Conference, San Diego, 1995.

[5] Allmaras S. Algebraic smoothing analysis of multigrid methods for the 2-D compressible Navier–Stokes equations. AIAA Paper 97-1954. In AIAA 13th Computational Fluid Dynamics Conference, Snowmass, 1997.

[6] Alonso JJ and Jameson A. Fully-implicit time-marching aeroelastic solutions. AIAA Paper 94-0056. In AIAA 32nd Aerospace Sciences Meeting, Reno, 1994.

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[10] Anderson BK, Thomas JL and Van Leer B. A comparison of flux vector splittings for the Euler equations. AIAA Paper 85-0122. In AIAA 23rd Aerospace Sciences Meeting, Reno, 1985.

Appendices

Times New Roman, Size 12, Header.

Times New Roman (Bold), Size 16

Times New Roman, Size 12.

Times New Roman (Bold, underlined), Size 14

Times New Roman (Bold, Italic), Size 12, Note: use appendix letters in numbering figure captions, i.e. A.1, A.2, B.1.

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Appendices

Appendices are not necessarily part of every thesis. Appendices may be used for supplementary illustrative material, original data, computer programs, and other material not necessarily appropriate for inclusion within the body text of the thesis.

Appendix A

Figure A.1 Shape optimization in additive manufacturing

Appendix B

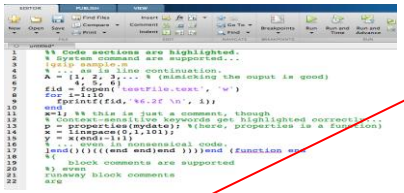

Figure B.1 Code excerpt

Appendix IV –Referencing style

ASME Referencing style

Text Citation. Within the text, references should be cited in numerical order according to their order of appearance. The numbered reference citation should be enclosed in brackets.

Example
It was shown by Prusa [1] that the width of the plume decreases under these conditions.

In the case of two citations, the numbers should be separated by a comma [1, 2]. In the case of more than two reference citations, the numbers should be separated by a dash [5-7].

List of References. References to original sources for cited material should be listed together at the end of the paper; footnotes should not be used for this purpose. References should be arranged in numerical order according to their order of appearance within the text.
(1) Reference to journal articles and papers in serial publications should include:
· last name of each author followed by their initials
· year of publication
· full title of the cited article in quotes, title capitalization
· full name of the publication in which it appears
· volume number (if any) in boldface (Do not include the abbreviation, "Vol.")
· issue number (if any) in parentheses (Do not include the abbreviation, “No.”)
· inclusive page numbers of the cited article (include “pp.”)
(2) Reference to textbooks and monographs should include:
· last name of each author followed by their initials
· year of publication
· full title of the publication in italics
· publisher
· city of publication
· inclusive page numbers of the work being cited (include “pp.”)
· chapter number (if any) at the end of the citation following the abbreviation, “Chap.”
(3) Reference to individual conference papers, papers in compiled conference proECEdings, or any other collection of works by numerous authors should include:
· last name of each author followed by their initials
· year of publication
· full title of the cited paper in quotes, title capitalization
· individual paper number (if any)
· full title of the publication in italics
· initials followed by last name of editors (if any), followed by the abbreviation, “eds.”
· publisher
· city of publication
· volume number (if any) in boldface if a single number, include, “Vol.” if part of larger identifier (e.g., “PVP-Vol. 254”)
· inclusive page numbers of the work being cited (include “pp.”)
(4) Reference to theses and technical reports should include:

· last name of each author followed by their initials
· year of publication
· full title in quotes, title capitalization
· report number (if any)
· publisher or institution name, city

Sample References

[1] Ning, X., and Lovell, M. R., 2002, "On the Sliding Friction Characteristics of Unidirectional Continuous FRP Composites," ASME J. Tribol., 124(1), pp. 5-13.
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[9] Smith, R., 2002, "Conformal Lubricated Contact of Cylindrical Surfaces Involved in a Non-Steady Motion," Ph.D. thesis, http://www.cas.phys.unm.edu/rsmith/homepage.html

Other referencing styles

You can use any established referencing style as long as you make sure that all referencing uses a single style. Elsevier journals offer the following referencing services and styles (the following example style refers to the journal of Computer Aided Design (CAD)):

Elsevier References

Citation in text: Please ensure that every reference cited in the text is also present in the reference list (and vice versa). Any references cited in the abstract must be given in full. Unpublished results and personal communications are not recommended in the reference list but may be mentioned in the text. If these references are included in the reference list, they should follow the standard reference style of the journal and should include a substitution of the publication date with either 'Unpublished results' or 'Personal communication'. Citation of a reference as 'in press' implies that the item has been accepted for publication.

Web references: As a minimum, the full URL should be given and the date when the reference was last accessed. Any further information, if known (DOI, author names, dates, reference to a source publication, etc.), should also be given. Web references can be listed separately (e.g., after the reference list) under a different heading, if desired, or can be included in the reference list.

References in a special issue: Please ensure that the words 'this issue' are added to any references in the list (and any citations in the text) to other articles in the same Special Issue.

Reference management software: Most Elsevier journals have a standard template available in key reference management packages. This covers packages using the Citation Style Language, such as Mendeley (http://www.mendeley.com/features/reference-manager) and also others like EndNote (http://www.endnote.com/support/enstyles.asp) and Reference Manager (http://refman.com/downloads/styles). Using plug-ins to word processing packages which are available from the above sites, authors only need to select the appropriate journal template when preparing their article and the list of references and citations to these will be formatted according to the journal style as described in this Guide. The process of including templates in these packages is constantly ongoing. If the journal you are looking for does not have a template available yet, please see the list of sample references and citations provided in this Guide to help you format these according to the journal style.

If you manage your research with Mendeley Desktop, you can easily install the reference style for this journal by clicking the link below: http://open.mendeley.com/use-citation-style/computer-aided-design when preparing your manuscript. You will then be able to select this style using the Mendeley plugins for Microsoft Word or LibreOffice. For more information about the Citation Style Language, visit http://citationstyles.org.

Reference style guidelines:

Text: Indicate references by number(s) in square brackets in line with the text. The actual authors can be referred to, but the reference number(s) must always be given.

List: Number the references (numbers in square brackets) in the list in the order in which they appear in the text.

Examples:

Reference to a journal publication:

[1] Van der Geer J, Hanraads JAJ, Lupton RA. The art of writing a scientific article. J Sci Commun 2010;163:51–9.

Reference to a book:

[2] Strunk Jr W, White EB. The elements of style. 4th ed. New York: Longman; 2000.

Reference to a chapter in an edited book:

[3] Mettam GR, Adams LB. How to prepare an electronic version of your article. In: Jones BS, Smith RZ, editors. Introduction to the electronic age, New York: E-Publishing Inc; 2009, p. 281–304.

Note shortened form for last page number. e.g., 51–9, and that for more than 6 authors the first 6 should be listed followed by 'et al.' For further details you are referred to 'Uniform Requirements for Manuscripts submitted to Biomedical Journals' (J Am Med Assoc 1997;277:927–34) (see also http://www.nlm.nih.gov/bsd/uniform_requirements.html).

Journal abbreviations source

Journal names should be abbreviated according to the List of Title Word Abbreviations: http://www.issn.org/services/online-services/access-to-the-ltwa/.

Appendix V –Declaration Form

The signed declaration form that follows should be first page after your report's cover sheet

DECLARATION

I hereby, declare that this manuscript, entitled *"title of thesis"*, is the result of my own work except for quotations and citations which have been duly acknowledged.
I also declare that, to the best of my knowledge and belief, it has not been previously or concurrently submitted, in whole or in part, for any other degree or diploma at Nazarbayev University or any other national or international institution.

(signature of author)

Name:
Date:

Appendix VI –Conflict of Interest

Definitions:

Conflict of interest - refers to a conflict between official University duties and private interests and personal relationships, where the private interests or personal relationships could improperly influence the way in which a person carries out their official duties.

Perceived conflict of interest - where a reasonable person might perceive that such improper influence as described above could exist.

Personal relationships - relationships with individuals or people that extend outside of the University or University duties, or a relationship where a reasonable person might perceive that there could be some bias, either positive or negative, resulting from that relationship. These include relationships with:

1. immediate family, e.g. spouse or partner, parents, children, step-children, etc.;
2. close relatives, e.g. aunts, uncles, cousins, nephews, nieces etc.;
3. rivals, e.g. competitors or persons with whom one has a history of serious conflict or enmity; and
4. all other relationships that could introduce bias in carrying out official duties.

Private interests - refers to any interests that involve potential gain or loss (financial or non-financial) for an individual or for any other person or organization that individual may wish to benefit (e.g. family, friends, associates) or disadvantage (e.g. competitors, rivals).

Rules and Guidelines:

1. All individuals are responsible for identifying, declaring, and managing conflicts of interest that apply to them.
2. Conflicts of interest may affect or have the appearance to affect sound and professional judgement adversely. Conflicts of interest or perceived conflicts of interest must be declared and managed to ensure integrity and transparency.
3. Staff members, students and other individuals who are charged with carrying out University activities and functions have a responsibility to declare and manage conflicts of interest as they arise. When declared, the conflict of interest should be avoided. Where this is not possible, action must be undertaken to ensure that the conflict (or perceived conflict) is managed in a transparent and appropriate manner.
4. Supervisors, chairs of committees/panels/groups and other responsible parties are accountable for ensuring that declared conflicts of interest, real or perceived, are evaluated and managed appropriately.
5. Declarations of conflict of interest should be made by individuals in writing to the relevant supervisor, chair of a committee/panel/group, other relevant person, or body as soon as the conflict is identified. Where circumstances prevent an immediate written declaration (e.g. conflict arises during a meeting), a verbal declaration should be lodged and, if possible, formally noted (e.g. in minutes of the meeting).

6. Where a supervisor, chair of a committee/panel/group, other relevant person or body becomes aware of a conflict of interest (or perceived conflict of interest) that has not been declared they should discuss the matter with the individual and take appropriate actions.
7. The best way in which to handle a conflict of interest is to avoid it. Where it is not possible to avoid a conflict of interest, the MSc ECE committee, the Head of Department, the School's Vice-Dean should be responsible for assessing the risk and taking appropriate actions.